

Comparativa de Semánticas Formales (WHILE Enriquecido)

1. Semántica Natural (NS - Big-Step)

$$\mathcal{T}_1 = \frac{\langle x := x + 1, \sigma_4 \rangle \longrightarrow \sigma_5 [\text{ass}_{\text{ns}}]}{\langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_4 \rangle \longrightarrow \sigma_5} [\text{repeat} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_2 = \frac{\langle x := x + 1, \sigma_3 \rangle \longrightarrow \sigma_4 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_1}{\langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_3 \rangle \longrightarrow \sigma_5} [\text{repeat} - \text{ff}_{\text{ns}}]$$

$$\mathcal{T}_3 = \frac{\langle x := x + 1, \sigma_2 \rangle \longrightarrow \sigma_3 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_2}{\langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_2 \rangle \longrightarrow \sigma_5} [\text{repeat} - \text{ff}_{\text{ns}}]$$

$$\mathcal{T}_4 = \frac{\langle x := x + 1, \sigma_1 \rangle \longrightarrow \sigma_2 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_3}{\langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_1 \rangle \longrightarrow \sigma_5} [\text{repeat} - \text{ff}_{\text{ns}}]$$

$$\mathcal{T}_5 = \frac{\langle x := x + 1, \sigma_0 \rangle \longrightarrow \sigma_1 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_4}{\langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_0 \rangle \longrightarrow \sigma_5} [\text{repeat} - \text{ff}_{\text{ns}}]$$

Estados

$$\sigma_0 = \{x \mapsto 0\}$$

$$\sigma_1 = \{x \mapsto 1\}$$

$$\sigma_2 = \{x \mapsto 2\}$$

$$\sigma_3 = \{x \mapsto 3\}$$

$$\sigma_4 = \{x \mapsto 4\}$$

$$\sigma_5 = \{x \mapsto 5\}$$

2. Semántica Operacional Estructurada (SOS - Small-Step)

$$\mathcal{T}_1 = \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_0 \rangle \Rightarrow \langle S_1, \sigma_0 \rangle [\text{repeat}_{\text{sos}}]$$

$$\mathcal{T}_2 = \frac{\langle x := x + 1, \sigma_0 \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle S_2, \sigma_0 \rangle \Rightarrow \langle S_3, \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_3 = \langle S_3, \sigma_1 \rangle \Rightarrow \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_1 \rangle [\text{if} - \text{ff}_{\text{sos}}]$$

$$\mathcal{T}_4 = \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_1 \rangle \Rightarrow \langle S_2, \sigma_1 \rangle [\text{repeat}_{\text{sos}}]$$

$$\mathcal{T}_5 = \frac{\langle x := x + 1, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]}{\langle S_1, \sigma_1 \rangle \Rightarrow \langle S_3, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_6 = \langle S_3, \sigma_2 \rangle \Rightarrow \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_2 \rangle [\text{if} - \text{ff}_{\text{sos}}]$$

$$\mathcal{T}_7 = \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_2 \rangle \Rightarrow \langle S_2, \sigma_2 \rangle [\text{repeat}_{\text{sos}}]$$

$$\mathcal{T}_8 = \frac{\langle x := x + 1, \sigma_2 \rangle \Rightarrow \sigma_3 [\text{ass}_{\text{sos}}]}{\langle S_2, \sigma_2 \rangle \Rightarrow \langle S_3, \sigma_3 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_9 = \langle S_3, \sigma_3 \rangle \Rightarrow \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_3 \rangle [\text{if} - \text{ff}_{\text{sos}}]$$

$$\mathcal{T}_{10} = \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_3 \rangle \Rightarrow \langle S_2, \sigma_3 \rangle [\text{repeat}_{\text{sos}}]$$

$$\mathcal{T}_{11} = \frac{\langle x := x + 1, \sigma_3 \rangle \Rightarrow \sigma_4 [\text{ass}_{\text{sos}}]}{\langle S_2, \sigma_3 \rangle \Rightarrow \langle S_3, \sigma_4 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{12} = \langle S_3, \sigma_4 \rangle \Rightarrow \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_4 \rangle [\text{if} - \text{ff}_{\text{sos}}]$$

$$\mathcal{T}_{13} = \langle \text{repeat } x := x + 1 \text{ until } x = 5, \sigma_4 \rangle \Rightarrow \langle S_2, \sigma_4 \rangle [\text{repeat}_{\text{sos}}]$$

$$\mathcal{T}_{14} = \frac{\langle x := x + 1, \sigma_4 \rangle \Rightarrow \sigma_5 [\text{ass}_{\text{sos}}]}{\langle S_2, \sigma_4 \rangle \Rightarrow \langle S_3, \sigma_5 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{15} = \langle S_3, \sigma_5 \rangle \Rightarrow \langle \text{skip}, \sigma_5 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{16} = \langle \text{skip}, \sigma_5 \rangle \Rightarrow \sigma_5 [\text{skip}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = (x := x + 1 ; \text{if } x = 5 \text{ then skip else repeat } x := x + 1 \text{ until } x = 5)$$

$$S_2 = (x := x + 1 ; \text{if } x = 5 \text{ then skip else repeat } x := x + 1 \text{ until } x = 5)$$

$$S_3 = \text{if } x = 5 \text{ then skip else repeat } x := x + 1 \text{ until } x = 5$$

Estados

$$\sigma_0 = \{x \mapsto 0\}$$

$$\sigma_1 = \{x \mapsto 1\}$$

$$\sigma_2 = \{x \mapsto 2\}$$

$$\sigma_3 = \{x \mapsto 3\}$$

$$\sigma_4 = \{x \mapsto 4\}$$

$$\sigma_5 = \{x \mapsto 5\}$$